Chapter Name

Environment and Sustainable development.

→ Environment

It refers to the total of all living things (biotic elements) and non-living things (abiotic elements) surrounding us.

Functions / Significance of Environment

1. Provide resources for production.
2. Environment assimilates waste.
3. Environment sustains life.
4. It provides aesthetic services.

Explanation —

(1) Provide Resources for Production

Any material gathered from the environment that is utilised by the society are environmental resources. This includes air for breathing, water, oil, metals, land for living and farming etc. Environmental resources can be renewable or non-renewable resources can be renewable or non-renewable. Renewable resources are those that can be renewed over time like water, air and light. Non-renewable resources like petroleum found under seas and oceans are resources that will be exhausted after continuous use. The environment continuously supplies its occupants with resources of all kinds.

(2) Environment Assimilates waste

The environment assimilates waste. That means environment absorbs the solid and liquid wastes created by humans and other livings.

(3) Environment sustains life

The environment sustains life on earth by providing both genetic and bio-diversities.

(4) It Provides Aesthetic Services

Environment also provides aesthetic services like beautiful scenery of forest, lands, water falls, mountain like the Himalya ranges, lagoons, glaciers, valleys etc.

→ Absorptive Capacity

(1) It refers to the capacity of the environment to absorb degradation.

(2) It is also knowns as assimilation capacity

→ Carrying Capacity

Carrying capacity implies two things :

(i) Resource extraction should remain below the rate of resource regeneration.

(ii) Generation of wastes should be remain within the absorption capacity of the environment

If these two conditions are not fulfilled, then an environmental crisis occurs.

→ Major Environmental Problem.

1. Global warming
2. Ozone layer depletion
3. Pollution
4. Land degradation.

→ Global Warming

It refers to a gradual increase in the overall temperature of the earth's atmosphere and oceans.

Causes of Global Warming

1. Global warming has caused due to increase in greenhouse gases like carbon dioxide, methane, ozone, nitrous oxide, water vapour etc.

2. The burning of coal and petroleum also increases greenhouse gases. Animal waste (animal fart) also leads to an increase in methane gas.

3. Deforestation increases the amount of carbon dioxide in the atmosphere.

Main effects of Global Warming

1. Ice is melting worldwide. It leads to a rising in sea level and floods in the coastal area.

2. Tropical storms are becoming stronger.

3. Increase in tropical diseases like malaria, Cholera, dengue, yellow fever, chikungunya etc.

4. So many species like polar bears, whales, etc. are in danger of becoming extinct forever.

→ Ozone Layer Depletion

1. It refers to the reduction in the amount of ozone in the stratosphere.
2. This problem is caused by a high level of CFCs used in cooling substances in air conditioners and refrigerators or BFCs used in fire extinguishers.
3. As a result of ozone layer depletion, more UV radiation comes to the earth causing damage to living organisms in the form of diseases like skin cancer.

→ Pollution

The major forms of pollution are as follows -

(i) Air Pollution
(ii) Water Pollution
(iii) Noise Pollution
(iv) Land Pollution

→ Land Degradation

It refers to a fall in the overall health of the land including the quality of soil and water commonly caused by human activities.

Factors Responsible for land degradation

(i) Loss of vegetation occuring due to defourstation

(ii) unsustainable fuel wood and fodder extraction

(iii) Shifting cultivation

(iv) encroachment into forest lands

(v) forest fires and over grazing

(vi) non- adoption of adequate soil conservation measures

(vii) Improper crop rotation

(viii) Indiscriminate use of agro- chemicals such as fertilisers and pesticides

(ix) Improper planning and management of irrigation systems

(x) extraction of ground water in excess of the recharge capacity.

(xi) open access resource

(xii) Poverty of the agriculture - dependent people.

→ Affluence Trap

1. With affluence or economic well-being more natural resources are used up causing environmental problems.

2. In the rich countries of the developed world, the high level of income is using natural resources at a fast speed.

→ India's Environmental Problem Pose a Dichotomy.

(1) Poverty is causing environmental degradation.
(2) Affluence in living standards is causing environmental degradation.

→ Sustainable Development

It refers to the process of development which meets the needs of the present generation without reducing the ability of future generations to meet their own needs.

→ Strategies for Sustainable Development.

1. Use of non-conventional sources of energy.
2. Use of cleaner fuels.
3. Establishment of mini-hydel plant
4. Traditional knowledge and practices.
5. Use of bio-compost
6. Control of biopest.
7. Change in unsustainable patterns of consumption and production.

Explanation :-

1) Use of non-conventional sources of energy -

India is hugely dependent on thermal and hydro power plants to meet its power needs both of these have adverse environmental impacts, thermal powerplants emit large quantities of carbon dioxide which is a green house gas. Hydroelectric project inundate forests and interfere with the natural flow of water in catchment areas and the river basins.

(2) LPG, Gobar gas in rural areas -

Households in rural areas generally use wood, dung cake or other biomass as fuel. This practice has several adverse implication like deforestation reduction in green cover, wastage cattle dung and air pollution. To rectify the situation subsidised LPG is provided. LPG is a clean fuel it reduces household pollution to a large extent. Also, energy wastage is minimised.

3) CNG in urban Areas -

In Delhi, the use of compressed natural gas (CNG) as fuel in public transport system has significally lowered airpout and the air has become cleaner in the last year.

4. Wind Power -

In areas where speed of wind is usually high. Windmills can provide electricity without any adverse impact on the environment. Wind turbines move with the wind and electricity is generated. No doubt, the initial cost is high. But the benefits are such that the high costs get easily absorbed.

5. Solar power through photovoice cells -

India is naturally endowed with a large quantity of solar energy in the form of solar energy in the form of sunlight. We use it in different way. e.g. we use sunlight. We use it to get the clothes, grains dried to keep our body warm in winter with the help of photovaVic cells. Solar energy can be converted into electricity These cells use speial kind of materials to capture solar energy and then convert the energy into electricity. This energy is extremely useful for remote area and for place where supply of power through guid or power lines is either not possible or powers very costly.

6. Mini - hydel plants -

Mountainous regions, stream can be found almost everywhere. A large percentage of such streams are perennial. Mini hydel plants use to energy of such streams to move smaU turbines the turbines generate electricity which can be used

locally, such power plants are more or less environment-friendly as they do not use pattern in areas where they are located, they generate enough power to meet local demands. This shows that they can reduce the requirement of large scale transmission tower and cables and avoid transmission loss.

7. Traditional Knowledge and Practices –

Traditionally, Indian people have been close to their environment. They have been more a component of environment and not its controller with the sudden onslaught of the western system of treatment. We were ignoring our traditional system such as Ayurveda, Unani, Tibetan and folk systems. Nowadays every cosmetic product, hair oil, toothpaste, body lotion, face cream and what not is herbal in composition, not only are these products environment friendly, they are relatively free from side effects and do not large scale industrial and chemical processing.

8. Biocomposting –

In our quest to increase agricultural production during the last five decades or so, we almost totally neglected the use of compost and completely switched to chemical fertilisers. The result is that large tracts of productive land have been adversely affected

water bodies including groundwater system have suffered due to chemical contamination and demand for irrigation has been going up year after year. In certain parts of the country, cattle are maintained only because they produce dung which is an important fertilizer and soil conditioner earthworms can convert organic matter into compost faster than the normal composting process. This process is now being widely used.

9- Bio pest Control -

With the advent of green revolution, the entire country entered into a frenzy to use more and more chemical pesticides for higher yield. Soon, the adverse impacts began to show food products were contaminated, soil, water bodies and even groundwater. Water was polluted with pesticides, even milk, meat and fishes were found to be contaminated. To meet his challenge efforts are on to bring better method of pest control with use of pesticides like neem are proving to be quite useful. Mixed cropping and growing different crops in consecutive years on the same land have also helped farmers.

→ Two instances of overuse of environmental resources are:

(i) excess use of electricity leads to depletion of resources like coal and water from which electricity is generated

(ii) Excess use of petrol and diesel in vehicles.

→ Two instances of misuse of environmental resources are:

(i) Soil degradation due to improper crop rotation and crop shifting.

(ii) Drying up of rivers due to supply of water from them to dams and reservoirs.

→ The Supply - demand reversal of environmental resources.

Before the advent of industrial revolution, the rate of resource extraction was less than the rate of regeneration of these resources. But with expanding population and its growing needs the demand for resources for both production and consumption went beyond the rate of regeneration of the resources. This has resulted in a reversal of supply - demand for relationship for environmental resources as now there is very high demand for environmental resources and services but their supply is limited due to overuse and misuse.

→ Central Pollution Control Board set up in 1974.

→ 17 industrial categories have been identified as significantly polluting by the CPCB.

→ Montreal Protocol - 1987

→ Environment Protection Act - 1986

→ Brundtland Commission - 1983
motive behind the formulation of the Brundtland Commission was -
1. to explore sustainable development and
2. the relationship between economic growth and environmental protection.